

Adaptive Entropy and Trust-Based DDoS Detection and Mitigation in MANETs

Final Year Project

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Abstract

Mobile Ad Hoc Networks (MANETs) are highly vulnerable to Distributed Denial of Service (DDoS) attacks due to their decentralized architecture, dynamic topology, and limited node resources. Such attacks can severely degrade network performance by overwhelming bandwidth, routing mechanisms, and node processing capabilities. This project proposes an Adaptive Entropy and Trust-Based Intrusion Detection and Mitigation System for mitigating DDoS attacks in MANET environments.

The proposed system integrates traffic-based detection, rate-based analysis, and entropy-based distributed attack detection to accurately identify malicious behavior. Upon detection, a trust-driven mitigation framework is activated, employing probabilistic rate limiting and temporary blacklisting to suppress malicious traffic while preserving legitimate communication. The system dynamically updates trust values based on node behavior, enabling adaptive prevention with minimal false positives.

The system is implemented and evaluated using the NS-3 simulator under realistic MANET scenarios with the AODV routing protocol. Performance metrics such as throughput, packet delivery ratio, and packet loss are analyzed to assess the effectiveness of the proposed approach. Simulation results demonstrate that the proposed system significantly improves network resilience under DDoS attacks, validating its suitability for secure and reliable MANET deployments.

Keywords: Mobile Ad Hoc Networks (MANETs), Distributed Denial of Service (DDoS), Intrusion Detection and Prevention, Entropy-Based Traffic Analysis, Trust-Based Mitigation, Adaptive Security Mechanisms, Network Traffic Anomaly Detection, AODV Routing Protocol, NS-3 Network Simulation

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Chapter 1

Introduction

1.1 Background/Overview

Mobile Ad Hoc Networks (MANETs) are decentralized wireless networks composed of self-configuring mobile nodes that communicate without relying on fixed infrastructure or centralized administration. Each node in a MANET functions both as a host and as a router, forwarding packets for other nodes to enable multi-hop communication. Due to their flexibility, rapid deployment, and self-organizing nature, MANETs are widely used in critical applications such as military communications, disaster recovery operations, emergency response systems, vehicular networks, and mobile Internet of Things (IoT) environments.

Despite these advantages, MANETs face significant security challenges. The absence of centralized control, open wireless medium, dynamic network topology, and reliance on node cooperation make them highly vulnerable to malicious attacks. Nodes in a MANET often operate under resource constraints, including limited bandwidth, processing power, and energy, which further complicates the implementation of traditional security mechanisms. As a result, ensuring secure and reliable communication in MANETs remains a major research challenge.

1.2 Problem Statement

Among various security threats targeting MANETs, Distributed Denial of Service attacks are particularly harmful. In a DDoS attack, malicious nodes overwhelm the network by generating excessive traffic, leading to congestion, increased packet loss, routing failures, and degradation of network performance. Such attacks can severely disrupt mission-critical MANET applications by denying legitimate users access to network services.

Existing security solutions often focus primarily on detecting abnormal traffic patterns associated with DDoS attacks. However, detection alone is insufficient to protect MANETs from sustained or adaptive attacks. Without an effective prevention mechanism, malicious traffic continues to consume network resources even after being identified. Furthermore, many traditional intrusion detection systems are unsuitable for MANET environments due to their centralized architecture, high computational overhead, and lack of adaptability to dynamic network conditions.

Therefore, there is a critical need for a **lightweight, adaptive, and distributed Intrusion Detection and Mitigation System** that can not only identify DDoS attacks in real time but also actively mitigate their impact while preserving legitimate network communication.

1.3 Aims and Objectives

The primary aim of this project is to design, implement, and evaluate an adaptive entropy and

trust-based Intrusion Detection and Mitigation System to mitigate DDOS attacks in Mobile Ad Hoc Networks. The proposed system analyzes MANET traffic behavior using packet count, rate-based, and entropy-based techniques to accurately identify malicious activity. To prevent attacks, a trust-driven mitigation mechanism is developed that applies probabilistic rate limiting and temporary blacklisting of malicious nodes. The detection and mitigation components are integrated into a unified, lightweight IDPS framework suitable for decentralized and dynamic MANET environments. The system is implemented using the NS-3 network simulator with the AODV routing protocol. Its effectiveness is evaluated using key performance metrics, including throughput, packet delivery ratio, packet loss, and overall network stability under attack conditions.

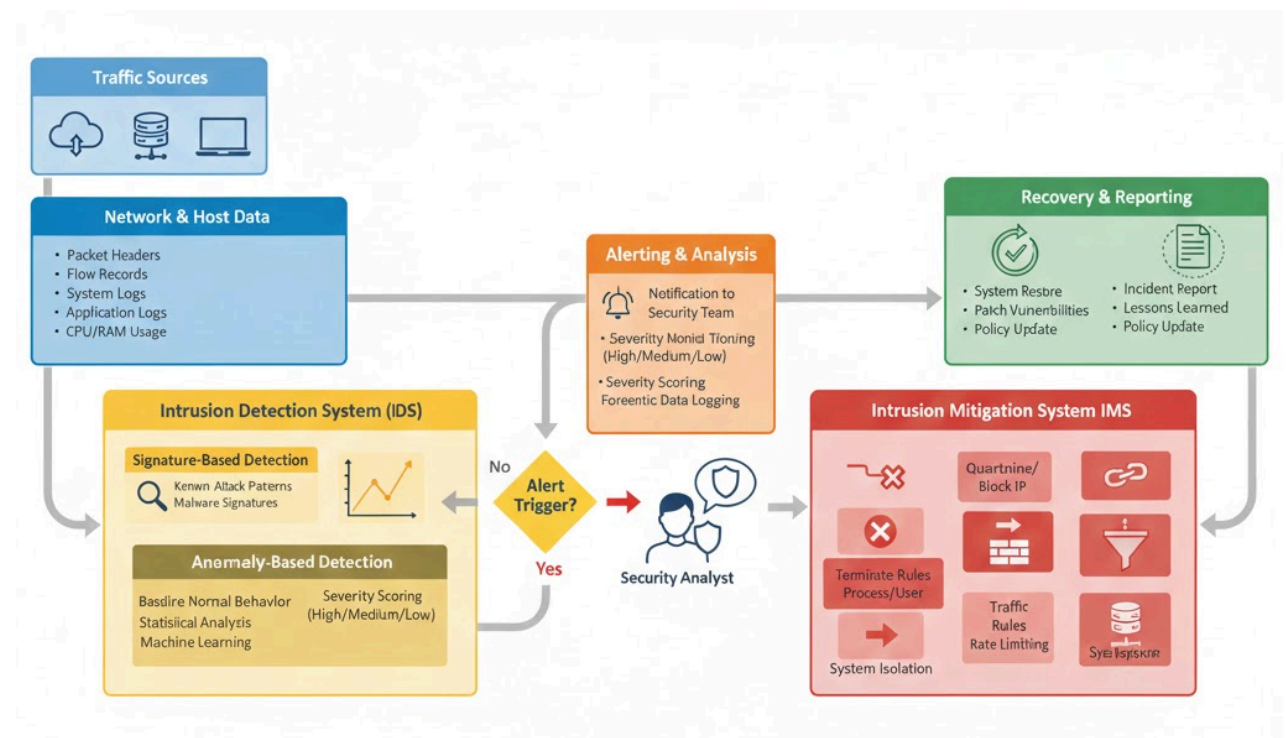


Fig.1 Intrusion Detection and Mitigation System Workflow

Chapter 2

Related Work

2.1 Overview of DDoS Attacks in MANETs

MANETs are highly vulnerable to Distributed Denial of Service (DDoS) attacks due to their decentralized architecture, open wireless medium, and reliance on node cooperation. Attackers exploit routing protocols and shared bandwidth to overwhelm network resources, leading to severe service degradation. Studies such as “*A Survey of DDoS Attacks in Mobile Ad Hoc Networks*” highlight that mobility and lack of centralized control significantly amplify attack impact.

Citation: “A Survey of DDoS Attacks in MANETs”, IEEE Communications Surveys

2.2 Traditional DDoS Detection Techniques

Traditional DDoS detection approaches include signature-based and threshold-based techniques. While signature-based methods are effective against known attacks, they fail to detect novel or adaptive threats. Threshold-based methods monitor traffic rate or packet volume but often suffer from false positives during normal traffic bursts, as discussed in “*Threshold-Based DDoS Detection in Wireless Networks*”.

Citation: “Threshold-Based DDoS Detection in Wireless Networks”, IEEE

2.3 Entropy-Based Traffic Analysis Approaches

Entropy-based approaches analyze randomness in traffic distributions to identify anomalies caused by DDoS attacks. These techniques are effective against distributed attack patterns and do not rely on predefined signatures. Research such as “*Entropy-Based Anomaly Detection for DDoS Attacks*” demonstrates improved detection accuracy in dynamic environments, though parameter tuning remains a challenge.

Citation: “Entropy-Based Anomaly Detection for DDoS Attacks”, Elsevier

2.4 Trust-Based Security Models in MANETs

Trust-based models assign trust values to nodes based on observed behavior, enabling identification and isolation of malicious participants. Such models are well-suited for decentralized MANETs, as shown in “*Trust-Based Security Framework for MANETs*”. However, improper trust recovery mechanisms may allow attackers to regain network access.

Citation: “Trust-Based Security Framework for MANETs”, IEEE Transactions

2.5 DDoS Mitigation Techniques in Ad Hoc Networks

DDoS mitigation strategies in MANETs include rate limiting, packet filtering, and node isolation. Recent studies emphasize adaptive and probabilistic mitigation to reduce false positives and maintain network connectivity. The paper “*Adaptive Mitigation of DDoS Attacks in Ad Hoc Networks*” highlights the importance of lightweight and distributed prevention mechanisms.

Citation: “Adaptive Mitigation of DDoS Attacks in Ad Hoc Networks”, Springer

2.6 Limitations of Existing Approaches

Existing approaches often focus solely on detection without integrating prevention, resulting in delayed or ineffective responses. Many solutions rely on static thresholds, centralized components, or high computational overhead, making them unsuitable for dynamic MANET environments. These limitations are discussed in “*Challenges in Securing MANETs Against DDoS Attacks*”.

Citation: “Challenges in Securing MANETs Against DDoS Attacks”, IEEE

2.7 Research Gap and Justification

The literature reveals a lack of integrated Intrusion Detection and Mitigation Systems that combine adaptive detection with effective mitigation in MANETs. While entropy-based detection and trust-based security models are individually explored, their unified application remains limited. This project addresses this gap by proposing an Adaptive Entropy and Trust-Based IDPS that provides both real-time detection and dynamic mitigation of DDoS attacks in MANETs.

Citation: “Integrated IDPS for DDoS Defense in MANETs”, IEEE

Chapter 3 Methodology

3.1 Research Methodology Overview

This research adopts a **simulation-based experimental methodology** to design, implement, and evaluate an adaptive Intrusion Detection and Prevention System for mitigating Distributed Denial of Service attacks in Mobile Ad Hoc Networks. The proposed methodology integrates **entropy-based traffic analysis for DDoS detection** with a **trust-based mitigation mechanism** to control malicious traffic behavior. The effectiveness of the proposed system is evaluated through extensive simulations conducted using the NS-3 network simulator under both normal and attack scenarios.

3.2 NS-3 Simulator Overview

NS-3 is an open-source, discrete-event network simulator widely used for research and academic experimentation in wireless and ad hoc networking. It provides realistic protocol implementations, flexible mobility models, and fine-grained control over network parameters. NS-3 is particularly suitable for MANET security research due to its support for routing protocols such as AODV and its ability to simulate large-scale attack scenarios. NS-3 is used to model network behavior, generate DDoS traffic, and evaluate the performance of the proposed system.

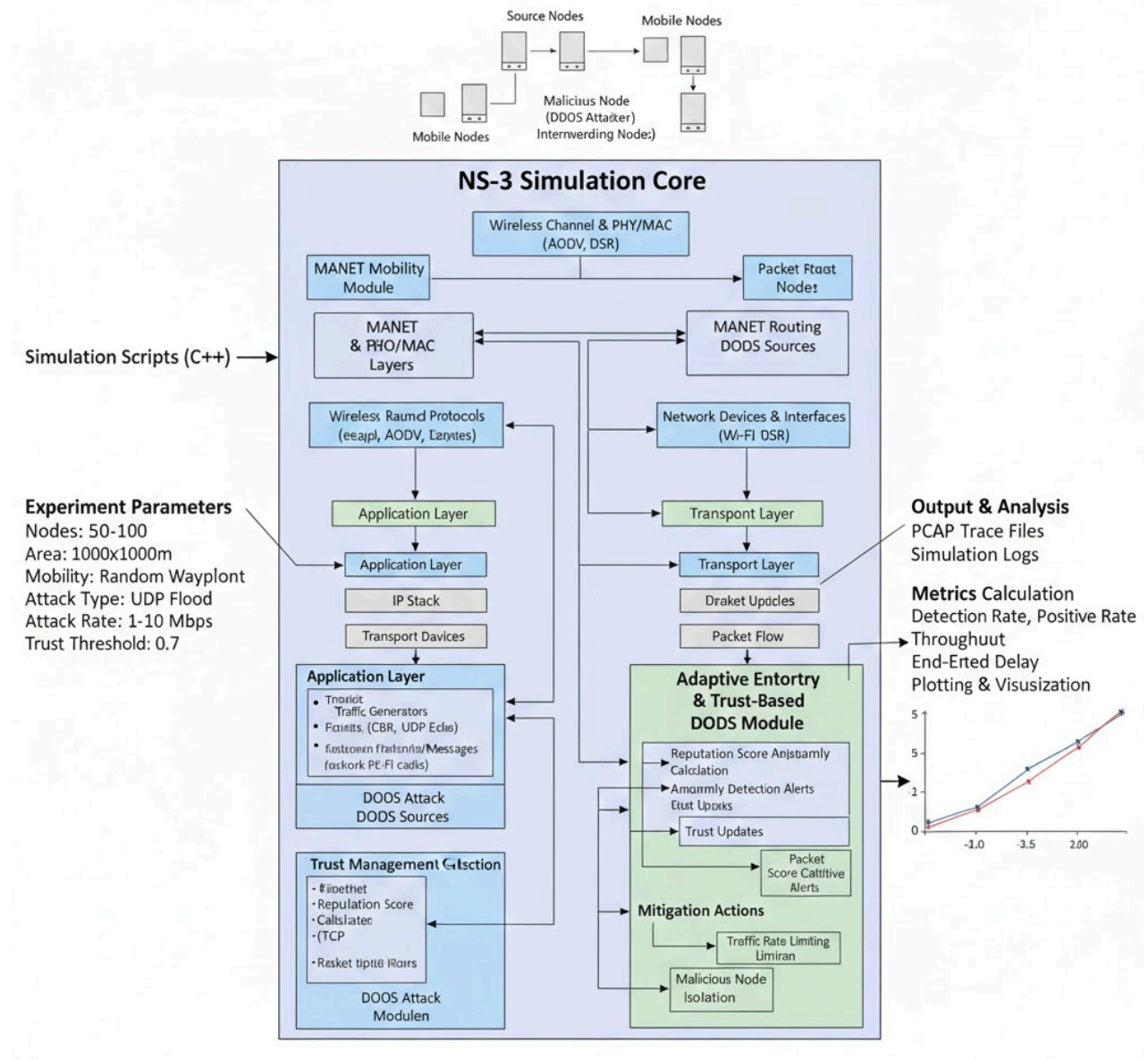


Fig.2 NS-3 Simulation Architecture DDOS defence for MANET

The **fig 2** show the whole NS3 simulation architecture of DDOS defence for mobile ad hoc network.

3.3 Network and Mobility Model Configuration

The simulated MANET environment consists of a variable number of mobile nodes deployed within a defined geographical area. Nodes communicate using IEEE 802.11 wireless standards and operate in an infrastructure-less configuration. Node mobility is modeled using the Random Waypoint Mobility Model to reflect realistic movement patterns. Simulation parameters such as node speed, pause time, transmission range, and simulation duration are carefully selected to evaluate the robustness of the proposed IDPS under dynamic network conditions.

3.4 Routing Protocol Selection (AODV-based MANET)

The Ad hoc On-Demand Distance Vector (AODV) routing protocol is selected due to its widespread use in MANET research and its vulnerability to DDoS attacks. AODV establishes routes on demand, reducing routing overhead but making it susceptible to flooding-based attacks. The proposed IDPS operates alongside AODV, monitoring routing and data traffic to identify abnormal behavior without modifying the core routing protocol, thereby preserving protocol compatibility.

3.5 Attack Model and Traffic Generation

To evaluate the proposed system, a DDoS attack model is implemented in which multiple malicious nodes generate high-rate traffic toward targeted victim nodes. Both legitimate and malicious traffic are generated using constant bit rate (CBR) and UDP-based flows. During attack scenarios, entropy-based analysis is applied to traffic features such as packet rate and source distribution to detect anomalies. Upon detection, the trust-based mitigation mechanism dynamically applies rate limiting and temporary blacklisting to suppress malicious traffic.

3.6 Performance Metrics

The performance of the proposed system is evaluated using standard MANET performance and security metrics.

3.6.1 Throughput

Throughput measures the total amount of successfully delivered data over the network per unit time. Higher throughput indicates improved network performance and effective mitigation of DDoS traffic.

3.6.2 Packet Delivery Ratio (PDR)

Packet Delivery Ratio is defined as the ratio of packets successfully received by destination nodes to the total packets transmitted by source nodes. A higher PDR reflects reliable communication and reduced attack impact.

3.6.3 Packet Loss

Packet loss represents the number of packets dropped during transmission due to congestion, attacks, or routing failures. Effective detection and mitigation mechanisms significantly reduce packet loss during DDoS attacks.

3.6.4 Detection and Mitigation Accuracy

Detection accuracy evaluates the ability of the IDPS to correctly identify malicious traffic, while mitigation accuracy measures the effectiveness of the prevention mechanism in suppressing attack traffic without affecting legitimate nodes. These metrics collectively assess the reliability and precision of the proposed IDPS.

Summary

This methodology enables a structured evaluation of an adaptive DDoS detection and mitigation system tailored for MANET environments, ensuring realistic simulation, measurable performance outcomes, and practical applicability.

Chapter 4

Adaptive DDoS Detection Framework

4.1 Overview of the Proposed Detection Architecture

The proposed adaptive DDoS detection framework is implemented as an **Intrusion Detection and Prevention System** within an ns-3-based Mobile Ad Hoc Network (MANET). The framework operates at the **transport layer (UDP socket reception point)** and performs **real-time traffic analysis** on incoming packets.

The architecture follows a **hybrid detection–mitigation model**, consisting of:

- **Multi-metric DDoS detection**
 - Packet count–based detection
 - Rate-based detection
 - Entropy-based distributed attack detection
- **Trust-based adaptive mitigation**
 - Probabilistic packet dropping
 - Temporary blacklisting of malicious nodes
- **Fully distributed operation**
 - Each node independently detects and mitigates attacks
 - No centralized controller or additional signaling overhead

Detection and mitigation decisions are executed in the `ReceivePacket()` function, ensuring immediate response to malicious traffic.

4.2 Packet Count–Based Flood Detection

Detection Principle

Packet count-based detection identifies flooding attacks by monitoring the **number of packets received from each source IP address** within an observation window. A node is suspected of launching a DDoS attack if it sends packets exceeding a predefined threshold.

Implementation Details

This mechanism is implemented in the CountDetection() function:

```
48      /* ===== Detection State ===== */
49      std::map<Ipv4Address, uint32_t> m_count;
```

- Each sender's packet count is tracked independently.
- A fixed threshold of **50 packets** is used.

```
114     bool
115     RoutingExperiment::CountDetection(Ipv4Address sender)
116     {
117         m_count[sender]++;
118         if (m_count[sender] > 50)
119         {
120             UpdateTrust(sender, -0.2);
121             m_count[sender] = 0;
```

Response Action

- The sender's trust value is reduced.
- The packet counter is reset to allow continued monitoring.
- Detection events are logged for analysis.

Effectiveness

This method is highly effective against:

- High-rate single-source flooding
- Resource exhaustion attacks targeting sink nodes

4.3 Rate-Based Flood Detection

Detection Principle

Rate-based detection focuses on the **temporal behavior of packet arrivals**, detecting abnormal packet rates even if packet counts remain moderate.

Implementation Details

The rate detection mechanism is implemented in RateDetection() using:

```
50     std::map<Ipv4Address, uint32_t> m_rate;  
51     std::map<Ipv4Address, Time> m_lastTime;  
52
```

- Packet arrivals are measured within a **1-second time window**.
- If more than **20 packets per second** are received from a sender, the node is flagged as malicious.

```
147  
148     if (m_rate[sender] > 20)  
149     {  
150         UpdateTrust(sender, -0.3);  
151         m_rate[sender] = 0;
```

Advantages

- Detects burst-based flooding attacks
- Resistant to attackers that evade static packet thresholds
- Complements packet count-based detection

4.4 Entropy-Based Distributed DDoS Detection

Detection Principle

Entropy-based detection identifies **distributed DDoS attacks**, where multiple nodes send traffic at low or moderate rates. The method evaluates the **randomness of source IP distribution** over time.

Mathematical Model

Entropy H is calculated as:

$$H = -\sum_{i=1}^n p_i \log_2(p_i)$$

- p_i is the probability of packets originating from sender i

Implementation Details

Implemented in the `EntropyDetection()` function using:

```
52
53     std::map<Ipv4Address, uint32_t> m_entropyCount;
54     Time m_entropyWindowStart{Seconds(0)};
55
```

- Observation window: **2 seconds**
- Entropy threshold: **$H > 3.5$**

```
183
184     if (H > 3.5)
185     {
186         UpdateTrust(sender, -0.4);
187         NS_LOG_UNCOND(now.GetSeconds())
```

Detection Capability

- Accurately detects **coordinated multi-node attacks**
- Effective against low-rate stealth DDoS traffic
- Minimal computational overhead due to window-based calculation

4.5 Detection Thresholds and Window Selection

Threshold Configuration

Detection Method	Threshold	Window Size
Packet Count	> 50 packets	Implicit
Rate-Based	> 20 pkt/s	1 second
Entropy-Based	$H > 3.5$	2 seconds

Table 1 . Detection Threshold and Window Selection

Design Rationale

- Thresholds were selected to balance **false positives and detection latency**
- Short windows enable rapid detection
- Longer entropy windows stabilize distributed traffic measurements

Adaptivity

The framework dynamically adapts via:

- Trust updates (UpdateTrust())
- Progressive mitigation escalation
- Automatic recovery for benign nodes

4.6 Detection Workflow and Decision Logic

1. Packet Reception

- UDP packet received at the sink node

2. Mitigation Pre-Check

- Blacklist and probabilistic drop applied

3. Multi-Layer Detection

- Packet count detection
- Rate-based detection
- Entropy-based detection

4. Trust Update

- Trust decreased for detected anomalies
- Trust slowly increased for normal traffic

5. Mitigation Decision

- Rate limiting
- Temporary blacklisting

6. Traffic Accounting

- Throughput and packet counters updated

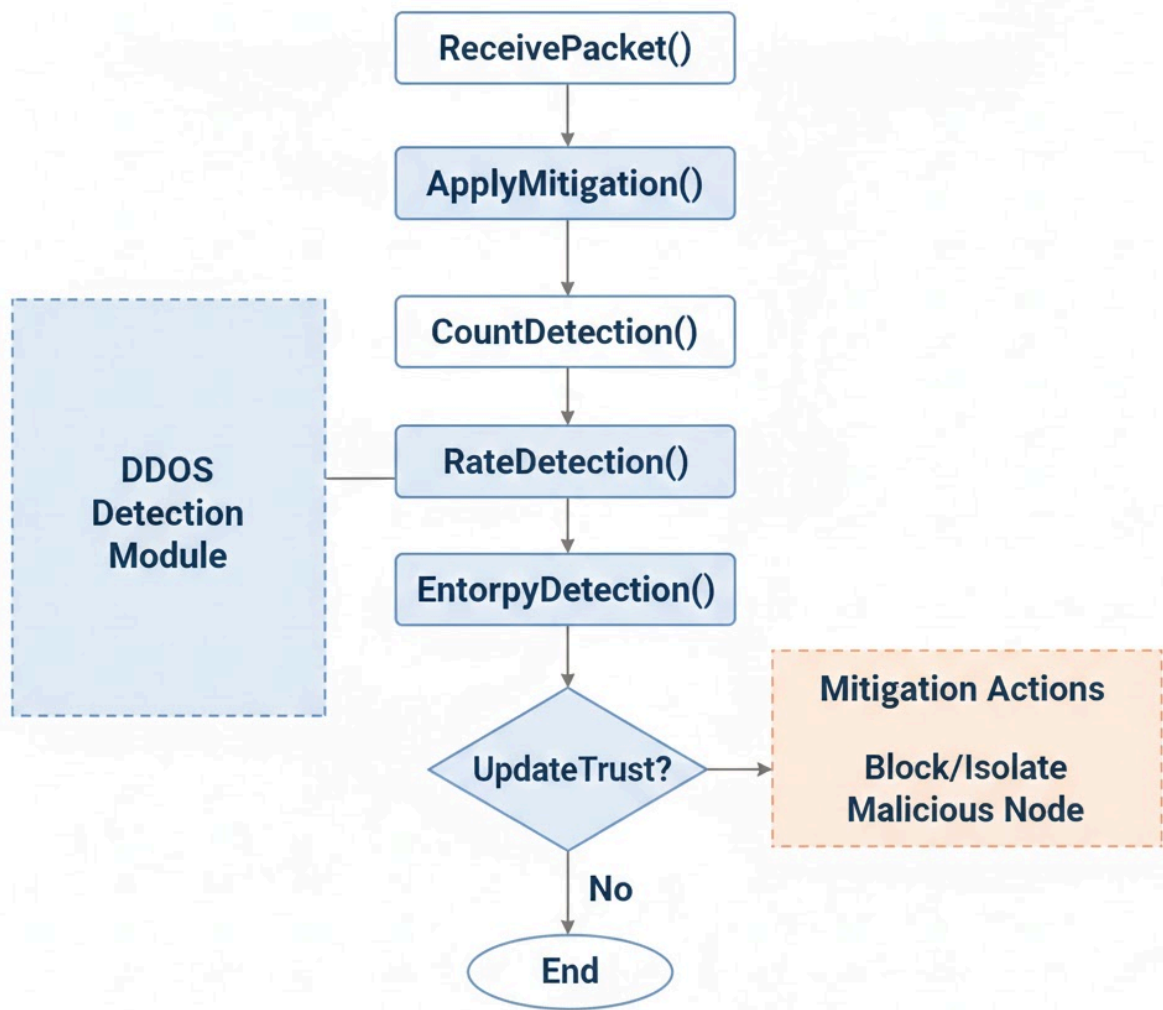


Fig.3 The Workflow Implementation of DDOS Detection Framework

Chapter 5

Trust-Based DDoS Mitigation Mechanism

5.1 Need for Mitigation in MANETs

Mobile Ad Hoc Networks (MANETs) are highly susceptible to DDoS attacks due to their decentralized nature, dynamic topology, and shared wireless medium. Malicious nodes can easily flood the network with excessive traffic, leading to bandwidth exhaustion, packet loss, and performance degradation. Conventional security solutions are often ineffective in MANET environments because of their high overhead and reliance on fixed infrastructure. Hence, there is a need for a lightweight and adaptive mitigation mechanism. A trust-based approach provides an effective solution by continuously evaluating node behavior and dynamically enforcing mitigation actions against malicious activity.

5.2 Overview of the Proposed Mitigation Strategy

The proposed mitigation strategy is integrated within the NS-3 simulation framework and operates alongside DDoS detection modules. Once suspicious behavior is detected, the system does not immediately isolate the node. Instead, it computes a **trust value** for each sender node based on its behavior over time. The mitigation strategy consists of:

- Trust computation and update
- Probabilistic rate-limiting
- Temporary blacklisting of malicious nodes
- Adaptive decision-making based on trust thresholds

This approach ensures fairness by allowing legitimate nodes to recover while strictly handling malicious nodes.

5.3 Trust Computation Model

The trust computation model assigns each node a trust score in the range $[0, 1]$, where:

- 1.0 indicates a fully trusted node
- 0.0 indicates a fully malicious node

Trust values are updated dynamically based on observed behavior.

5.3.1 Trust Initialization

When a node communicates for the first time, it is assigned a default trust value of **1.0**, assuming benign behavior unless proven otherwise.

```
95     {  
96         if (!m_trust.count(sender))  
97             m_trust[sender] = 1.0;  
98     }
```

This optimistic initialization avoids penalizing new or infrequent nodes.

5.3.2 Trust Update Based on Node Behavior

Trust is adjusted using positive or negative deltas depending on detection outcomes. Negative deltas are applied when DDoS behavior is detected, while positive deltas are applied during normal traffic flow.

```
98  
99     m_trust[sender] = std::clamp(m_trust[sender] + delta, 0.0, 1.0);  
100
```

Different detection modules impose different penalties:

- Count-based flooding: -0.2
- Rate-based flooding: -0.3
- Entropy-based DDoS: -0.4

5.4 Probabilistic Rate-Limiting Mechanism

Instead of completely blocking nodes immediately, the system applies **probabilistic packet dropping** based on trust levels. Nodes with moderate trust are partially restricted.

```
80  /* Probabilistic drop */
81  double prob = m_dropProb[sender];
82  if (m_rng->GetValue() < prob)
83  {
84      NS_LOG_UNCOND(now.GetSeconds()
85                  << " [MITIGATION] Rate-limited packet dropped from "
86                  << sender);
87      return true;
88  }
89
90  return false;
91 }
```

This mechanism reduces attack impact while minimizing false positives.

5.5 Blacklisting of Malicious Nodes

Nodes whose trust value drops below a critical threshold are temporarily blacklisted. During the blacklist duration, all packets from the node are dropped.

```
106 {
107     m_blacklist[sender] = Simulator::Now() + Seconds(20);
108     m_dropProb[sender] = 1.0;
109 }
```

Blacklist entries are time-bound, allowing nodes to rejoin the network after isolation.

5.6 Adaptive Mitigation Decision Process

Mitigation decisions are adaptive and depend on the current trust value:

Trust Range	Mitigation Action
≥ 0.7	No restriction
0.4 – 0.69	50% packet drop
< 0.4	Blacklisting


```

100
101     if (m_trust[sender] >= 0.7)
102     |     m_dropProb[sender] = 0.0;
103     else if (m_trust[sender] >= 0.4)
104     |     m_dropProb[sender] = 0.5;
105     else
106     |

```

This ensures proportional response based on severity.

5.7 Integration of Detection and Mitigation Modules

The mitigation mechanism is tightly integrated with detection modules. Detection is performed first, followed by mitigation decisions before packet processing.

```

209     /* Apply mitigation BEFORE processing */
210     if (ApplyMitigation(sender))
211     |     continue;

```

Trust is then updated based on detection outcomes and normal traffic behavior.

5.8 Mitigation Algorithm Flow

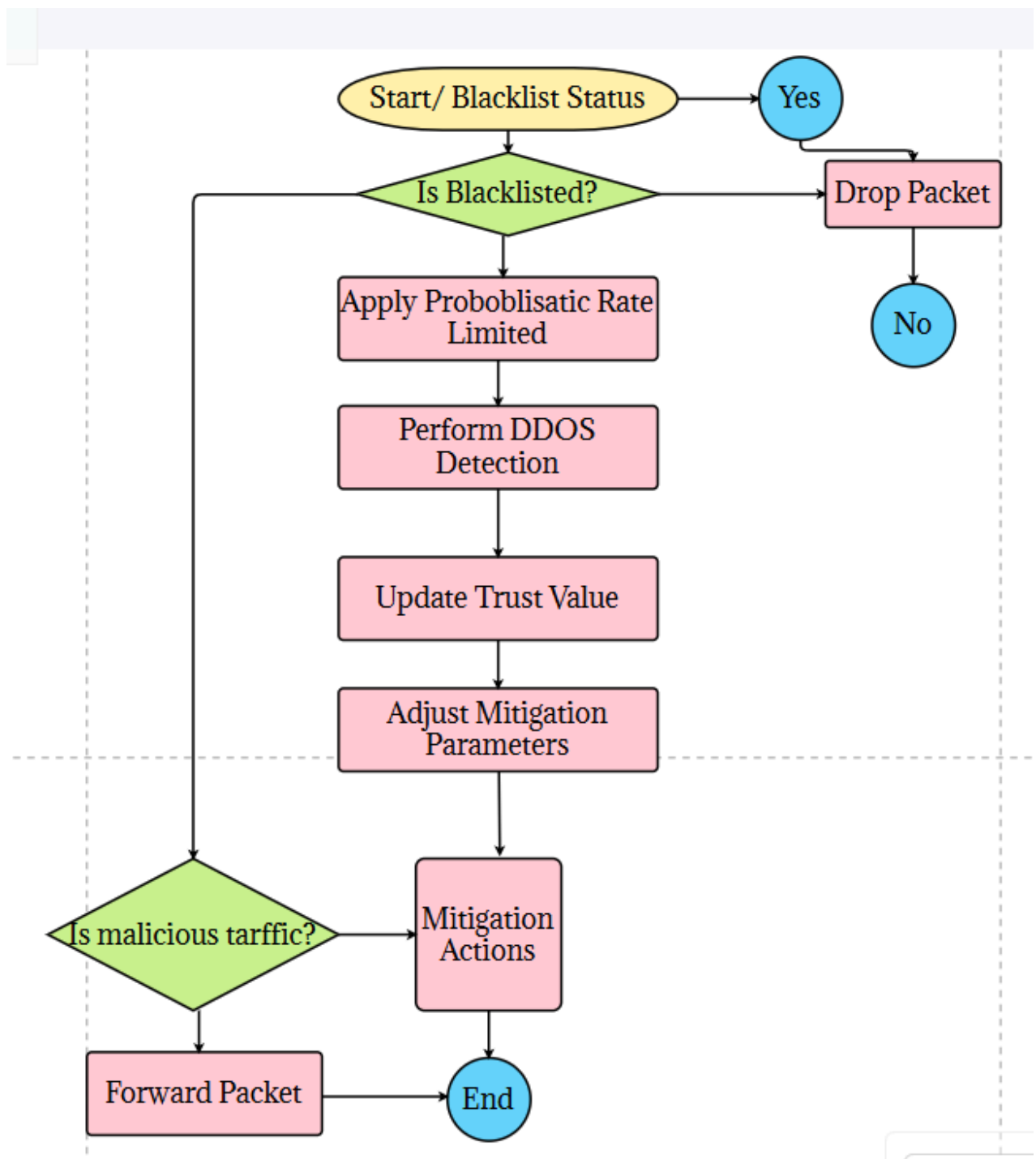


Fig.4 The Flow of Mitigation Algorithm

The overall mitigation process ensures real-time adoptive defense. Packets are first checked against a blacklist, then rate-limiting. The shapes used in **Figure 4** represent Ovals (Terminators), Rectangles (Processes), and Diamonds (Decisions).

Chapter 6

System Implementation

6.1 Overall System Architecture

The proposed system is implemented in the NS-3 network simulator and follows a **modular intrusion detection and prevention architecture** tailored for MANET environments. The architecture consists of three primary components: the traffic generation module, the detection module, and the trust-based mitigation module. These components operate in conjunction with the AODV routing protocol to monitor, detect, and mitigate DDoS attacks in real time.

Incoming packets are first subjected to mitigation checks. If permitted, they are then analyzed by multiple detection mechanisms. Based on detection outcomes, node trust values are dynamically updated, influencing future mitigation decisions. This layered design ensures adaptive and efficient DDoS defense. The **fig 5** show the overall working of DDOS detection and Mitigation for MANETs

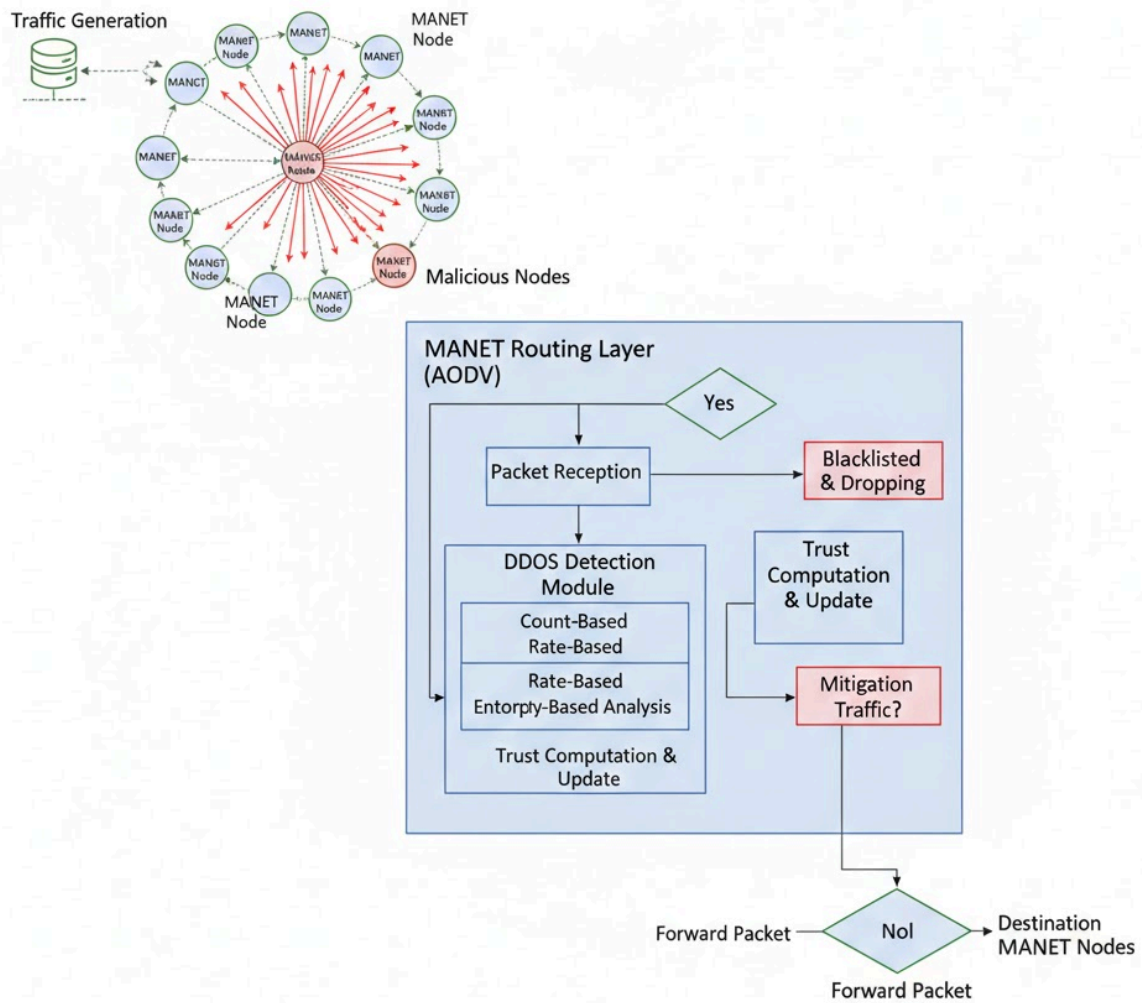


Fig.5 Overall System Architecture

6.2 Code Structure and Module Description

The system is implemented within a single C++ class named `RoutingExperiment`. This class encapsulates all major functionalities, including packet reception, detection logic, trust management, and mitigation enforcement.

- **Packet Reception Module:** Handles incoming traffic using UDP sockets.
- **Detection Module:** Identifies abnormal traffic patterns using count-based, rate-based, and entropy-based techniques.
- **Mitigation Module:** Applies probabilistic rate limiting and blacklisting based on trust values.

- **Routing Module:** Uses the AODV routing protocol for multi-hop communication.
- **Monitoring Module:** Logs detection and mitigation events.

The modular design improves readability, maintainability, and extensibility of the system.

6.3 Detection Module Implementation

The detection module continuously monitors incoming packets to identify potential DDoS behavior. Three complementary detection techniques are implemented:

1. **Count-Based Detection:** Detects flooding by monitoring the number of packets received from a sender within a threshold.
2. **Rate-Based Detection:** Measures packet arrival rates per second to identify high-speed flooding.
3. **Entropy-Based Detection:** Calculates entropy to detect distributed attack patterns from multiple sources.

Each detection method independently updates the trust value of the sender upon detecting malicious behavior. This multi-metric approach improves detection accuracy and robustness.

6.4 Mitigation Module Implementation

The mitigation module is responsible for enforcing traffic control decisions based on node trust values. It is executed **before packet processing**, ensuring that malicious traffic is filtered early.

6.4.1 ApplyMitigation() Function

The ApplyMitigation() function determines whether a packet should be dropped. It first checks whether the sender is blacklisted. If not, it applies probabilistic rate limiting based on the sender's drop probability. This approach enables gradual mitigation rather than immediate isolation, reducing false positives.

6.4.2 Trust Update Logic

Trust values are updated dynamically using both positive and negative reinforcement. Negative trust updates are applied when detection modules identify suspicious behavior, while positive updates are applied during normal packet reception. Trust values are bounded within the range $[0,1]$ to maintain stability. Based on trust thresholds, appropriate mitigation actions are selected.

6.4.3 Blacklisting and Rate Limiting

Nodes with critically low trust values are temporarily blacklisted, causing all packets from them to be dropped for a predefined duration. Nodes with moderate trust values are subjected to probabilistic rate limiting. This tiered response ensures proportional mitigation while allowing legitimate nodes to recover over time.

6.5 Integration with AODV Routing Protocol

The system integrates seamlessly with the AODV routing protocol provided by NS-3. AODV is selected due to its on-demand route discovery and suitability for dynamic MANET environments. The detection and mitigation logic operates at the application layer and does not modify AODV's internal mechanisms. This design preserves routing stability while enhancing security against DDoS attacks.

6.6 Simulation Parameters and Configuration

The simulation environment consists of 50 mobile nodes deployed in an ad hoc wireless network. Nodes communicate using IEEE 802.11b standard with a constant data rate.

Random Waypoint Mobility Model is used to simulate realistic node movement. Traffic is generated using UDP On-Off applications, and the simulation runs for 200 seconds. These parameters are chosen to reflect a dynamic and realistic MANET scenario.

6.7 Logging and Monitoring Mechanisms

The system uses NS-3 logging utilities to record detection and mitigation events. Logs include alerts for count-based flooding, rate-based flooding, entropy-based DDoS detection, packet drops, and blacklist enforcement.

Throughput monitoring is performed periodically to evaluate network performance. These logs enable detailed post-simulation analysis of attack impact and mitigation effectiveness.

Chapter 7

Results and Performance Evaluation

7.1 Simulation Scenarios

Evaluate the effectiveness of the proposed Adaptive Entropy Trust-Based DDoS Detection and Mitigation mechanism, multiple simulation scenarios were designed using the NS-3 network simulator. The simulations were conducted in a MANET environment consisting of 50 mobile nodes using the AODV routing protocol. IEEE 802.11b was used as the wireless communication standard, and node mobility followed the Random Waypoint Mobility Model.

Three primary scenarios were considered:

1. Normal network operation without any attack
2. Network under DDoS attack without mitigation
3. Network under DDoS attack with the proposed mitigation enabled

These scenarios allow a fair comparison of network performance before and after applying the proposed mechanism.

7.2 Network Performance Without Mitigation

In the absence of any attack, the network operates under normal conditions where all nodes cooperate in packet forwarding. Traffic is generated using UDP On-Off applications, and no malicious behavior is introduced.

```
muna@muna-Lenovo-V14-G3-IAP: ~/Downloads/ns-allinone-3.43/ns-3.43
140.757 0 received one packet from 10.1.1.19
140.875 0 received one packet from 10.1.1.11
140.907 7 received one packet from 10.1.1.18
140.985 2 received one packet from 10.1.1.13
140.988 2 received one packet from 10.1.1.13
140.991 2 received one packet from 10.1.1.13
141.009 8 received one packet from 10.1.1.19
141.116 0 received one packet from 10.1.1.11
141.157 7 received one packet from 10.1.1.18
141.257 8 received one packet from 10.1.1.19
141.386 0 received one packet from 10.1.1.11
141.407 7 received one packet from 10.1.1.18
141.509 2 received one packet from 10.1.1.13
```

Under this scenario, the network demonstrates stable throughput, low packet loss, and a high packet delivery ratio. This baseline performance is used as a reference to evaluate the impact of DDoS attacks and the effectiveness of mitigation techniques.

7.3 Impact of DDoS Attack on MANET Performance

When a DDoS attack is introduced without any mitigation mechanism, the network performance degrades significantly. Malicious nodes generate excessive traffic, overwhelming the destination node and consuming network bandwidth.

- Packet collisions increase
- Legitimate packets are dropped
- Throughput fluctuates and decreases
- Packet Delivery Ratio drops sharply


```
muna@muna-Lenovo-V14-G3-IAP: ~/Downloads/ns-allinone-3.43/ns-3.43
112.889 - Received packet from 10.1.1.17
112.896 - Received packet from 10.1.1.18
112.934 - Received packet from 10.1.1.12
112.97 - Received packet from 10.1.1.15
113.001 - Received packet from 10.1.1.19
113.035 - Received packet from 10.1.1.14
113.186 - Received packet from 10.1.1.12
113.22 - DDoS Attack Detected! Flood from 10.1.1.15. Dropping packets.
113.255 - Received packet from 10.1.1.19
113.285 - Received packet from 10.1.1.14
113.405 - Received packet from 10.1.1.18
113.434 - DDoS Attack Detected! Flood from 10.1.1.12. Dropping packets.
113.47 - Received packet from 10.1.1.15
113.509 - Received packet from 10.1.1.19
113.535 - Received packet from 10.1.1.14
113.646 - Received packet from 10.1.1.18
113.684 - Received packet from 10.1.1.12
113.72 - Received packet from 10.1.1.15
113.749 - Received packet from 10.1.1.19
113.785 - Received packet from 10.1.1.14
113.896 - Received packet from 10.1.1.18
113.934 - Received packet from 10.1.1.12
113.97 - Received packet from 10.1.1.15
113.999 - Received packet from 10.1.1.19
114.043 - DDoS Attack Detected! Flood from 10.1.1.14. Dropping packets.
114.146 - Received packet from 10.1.1.18
114.184 - Received packet from 10.1.1.12
114.22 - Received packet from 10.1.1.15
114.257 - Received packet from 10.1.1.19
114.285 - Received packet from 10.1.1.14
114.396 - Received packet from 10.1.1.18
114.434 - Received packet from 10.1.1.12
114.47 - Received packet from 10.1.1.15
114.535 - Received packet from 10.1.1.14
```

This scenario highlights the vulnerability of MANETs to DDoS attacks and emphasizes the need for an effective detection and mitigation strategy.

7.4 Performance After Applying Mitigation

After enabling the proposed adaptive entropy and trust-based mitigation mechanism, a noticeable improvement in network performance is observed. Malicious traffic is gradually restricted through probabilistic rate limiting and blacklisting based on trust values.

Legitimate nodes are allowed to recover over time due to trust reinforcement, ensuring fairness. The network stabilizes as attack traffic is suppressed, leading to improved throughput and packet delivery.

7.5 Packet Delivery Ratio (PDR) Analysis

Packet Delivery Ratio (PDR) is defined as the ratio of successfully received packets to the total number of packets sent by the source nodes.

With mitigation enabled, the PDR shows a significant improvement compared to the attack-only scenario. This indicates that the proposed mechanism effectively protects legitimate traffic while isolating malicious nodes.

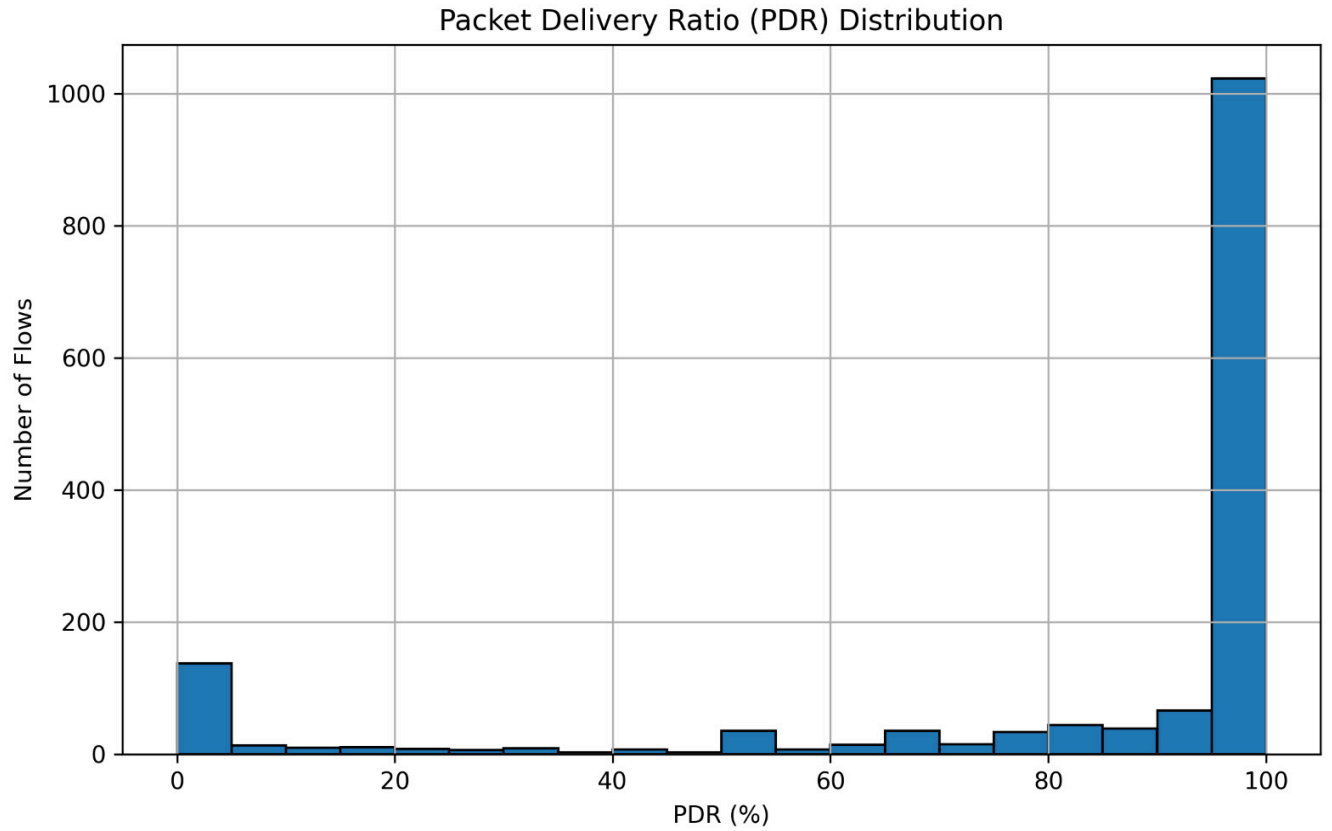


Fig. 6 Packet Delivery Ratio comparison with and without mitigation

The **Fig 6** demonstrates that PDR remains high during normal operation, drops sharply during DDoS attacks, and recovers substantially after applying the mitigation mechanism.

7.6 Throughput Analysis Over Time

Throughput represents the rate of successful data delivery over the network, measured in bits per second. During DDoS attacks without mitigation, throughput becomes unstable due to excessive malicious traffic. After enabling the proposed mitigation mechanism, throughput stabilizes as attack traffic is controlled and legitimate traffic is prioritized.

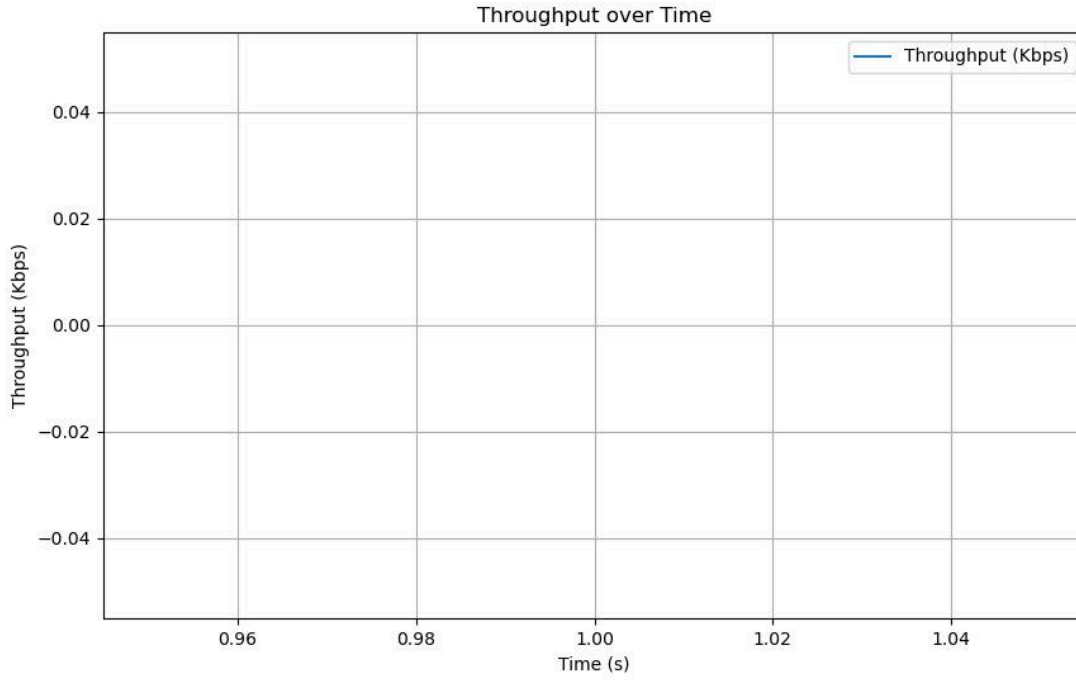


Fig. 7 Throughput variation over time

The **Fig 7** show the results confirm that the adaptive trust-based mitigation improves overall network efficiency.

7.7 Detection vs Mitigation Effectiveness Comparison

Detection alone is insufficient to protect the network, as identifying malicious behavior does not prevent continued attack traffic. The proposed system combines detection with adaptive mitigation, resulting in better performance recovery.

Entropy-based detection effectively identifies distributed DDoS attacks, while trust-based mitigation ensures proportional response. The combined approach achieves higher PDR and throughput compared to detection-only scenarios.

7.8 Discussion of Results

The simulation results demonstrate that the proposed **Adaptive Entropy Trust-Based DDoS Detection and Mitigation mechanism** significantly enhances MANET performance under attack conditions. The trust-based approach reduces false positives by allowing gradual response rather than immediate isolation.

Entropy-based detection proves effective in identifying distributed attack patterns, while adaptive mitigation ensures network stability and fairness. Overall, the results validate that the proposed mechanism provides an efficient, scalable, and lightweight solution for securing MANETs against DDoS attacks.

Chapter 8

Visualization and Analysis

8.1 Wireshark Traffic Analysis

The Wireshark I/O graph is based on a packet capture file (merge.pcap). This graph (**Fig 8**) shows the trend of packets per second (pps) over time (in seconds).

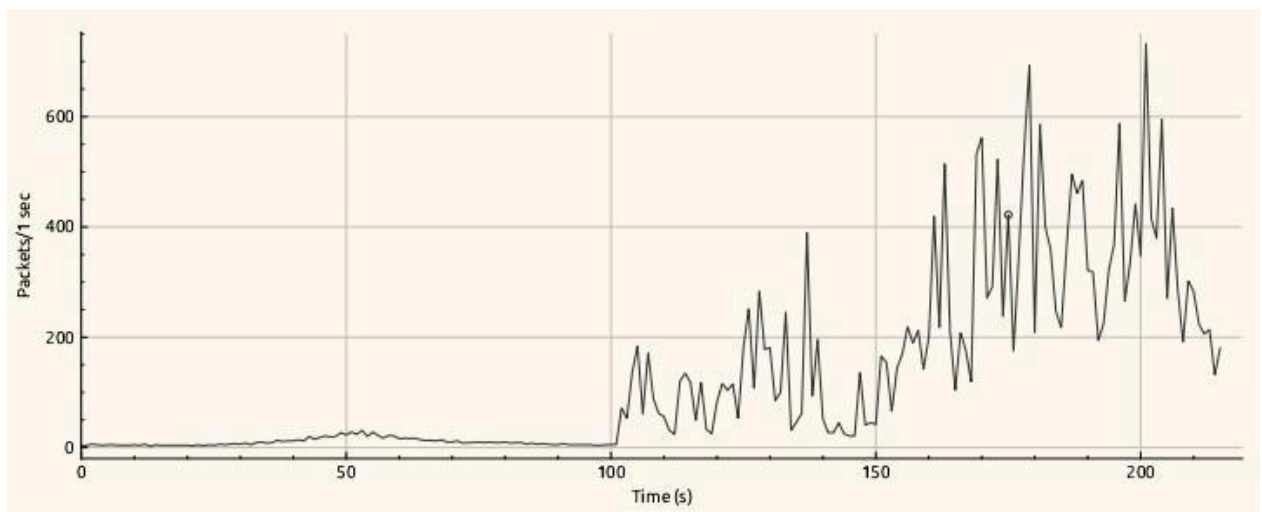
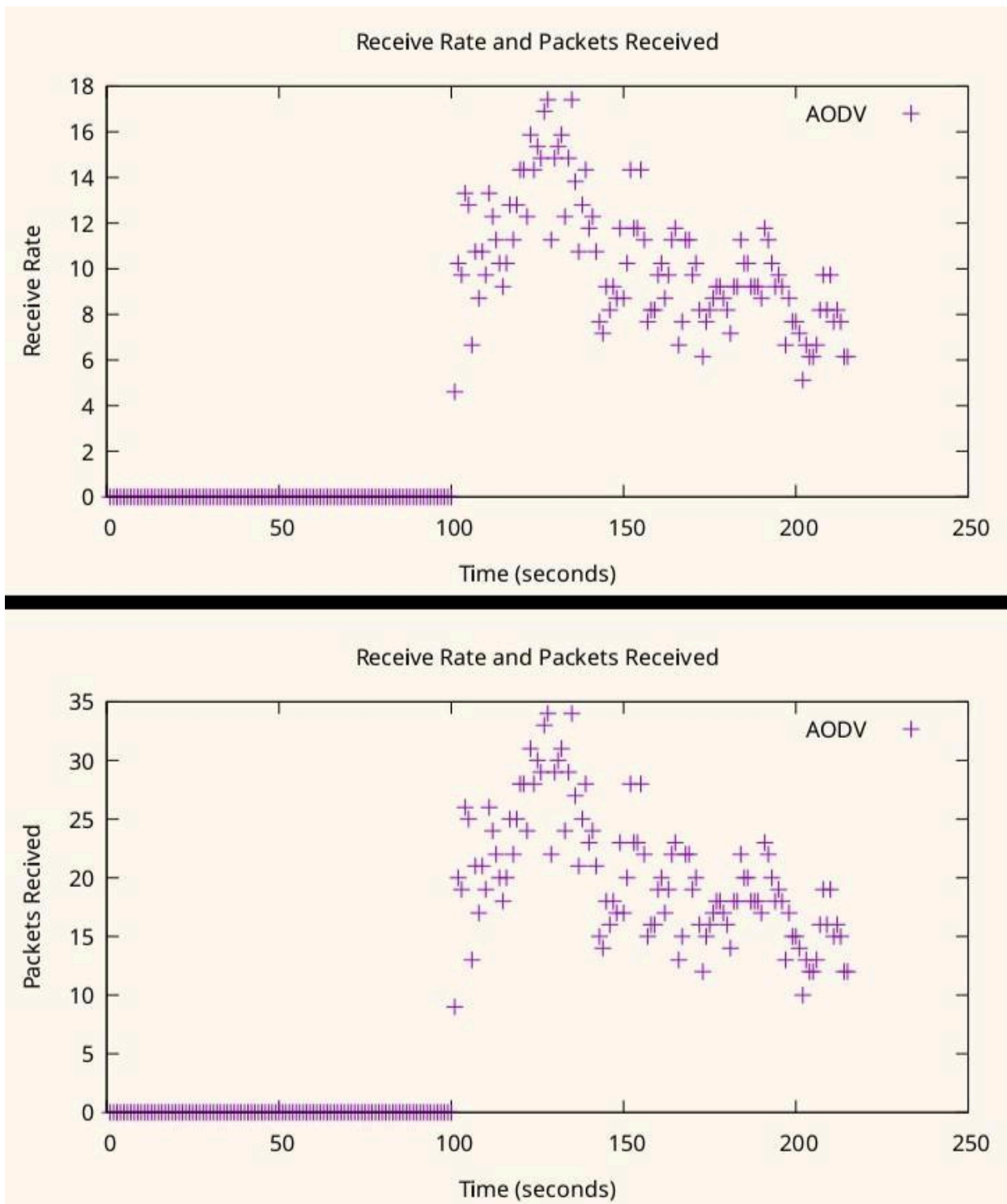


Fig. 8 Wireshark I/O graph for the simulation

The **image (Fig 9)** shows two graphs analyzing the *AODV protocol* in MANETs, focusing on **Receive Rate** and **Packets Received** over time.



The graphs show AODV protocol performance in a dynamic network, with fluctuating packet receipt after initial route establishment.

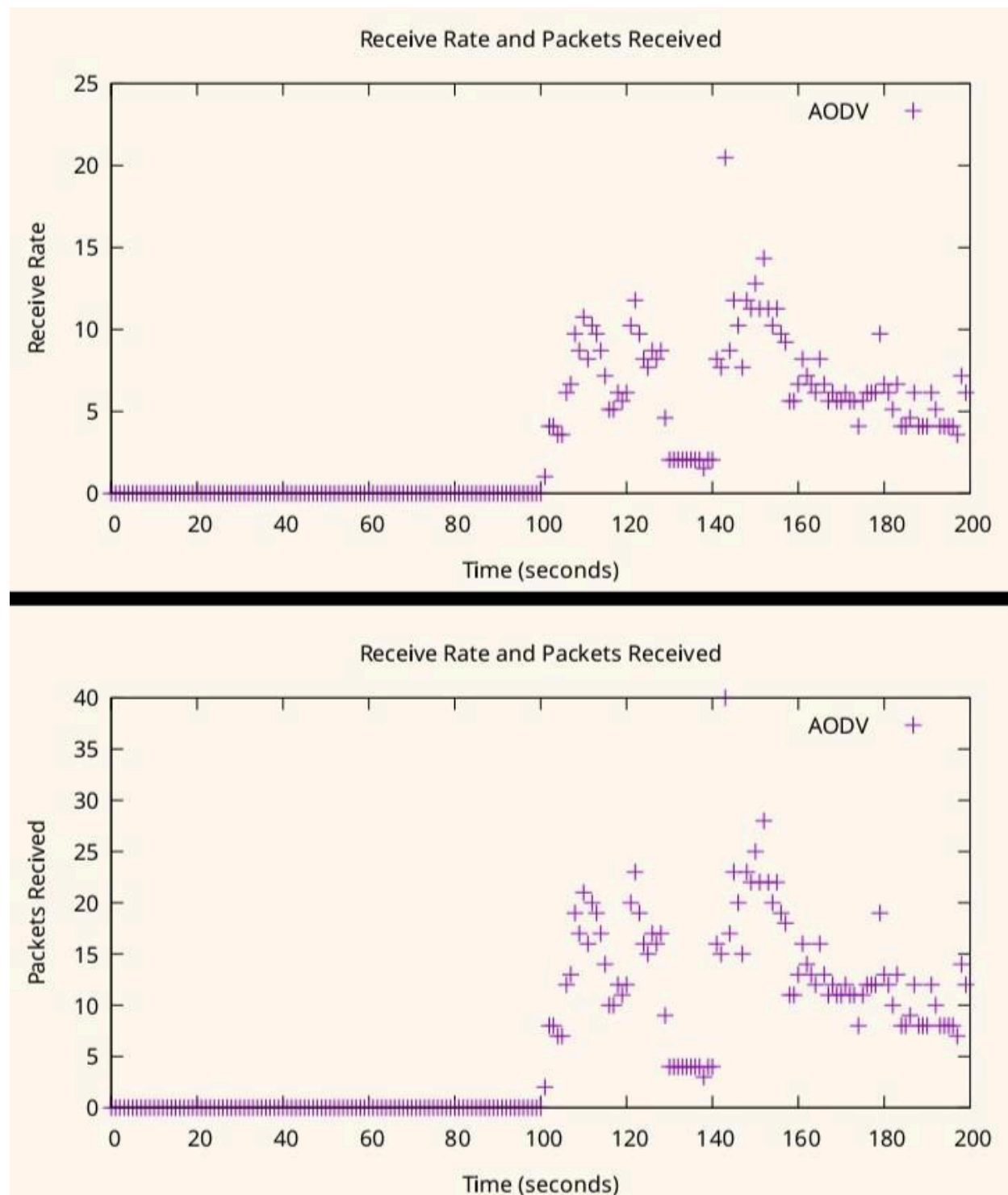


Fig 9. Analyzing the *AODV* protocol in MANETs

8.2 FlowMonitor Statistics Evaluation

The **figure 10** shows the flow-monitor statistics from a network simulation, displaying details such as transmission and reception bitrates, mean delay, packet loss ratios, and packets dropped for various flow IDs. These metrics are useful for analyzing network performance and detecting anomalies such as potential DDoS attacks through traffic analysis.

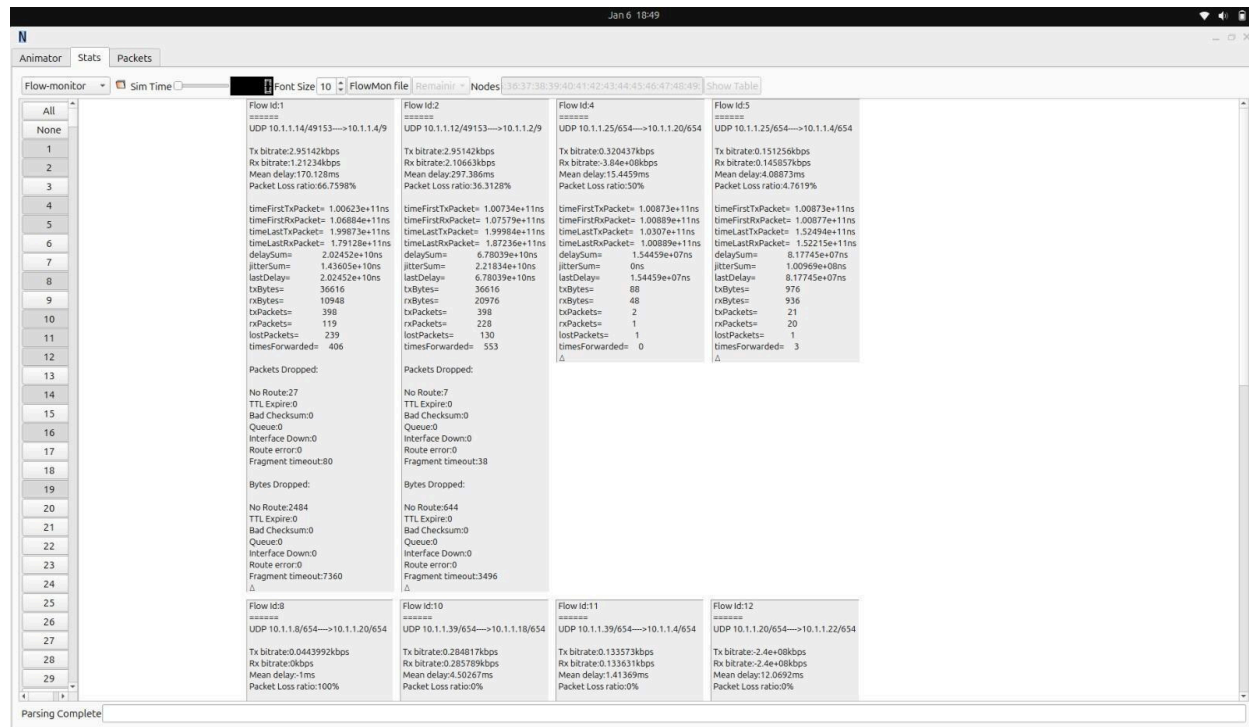
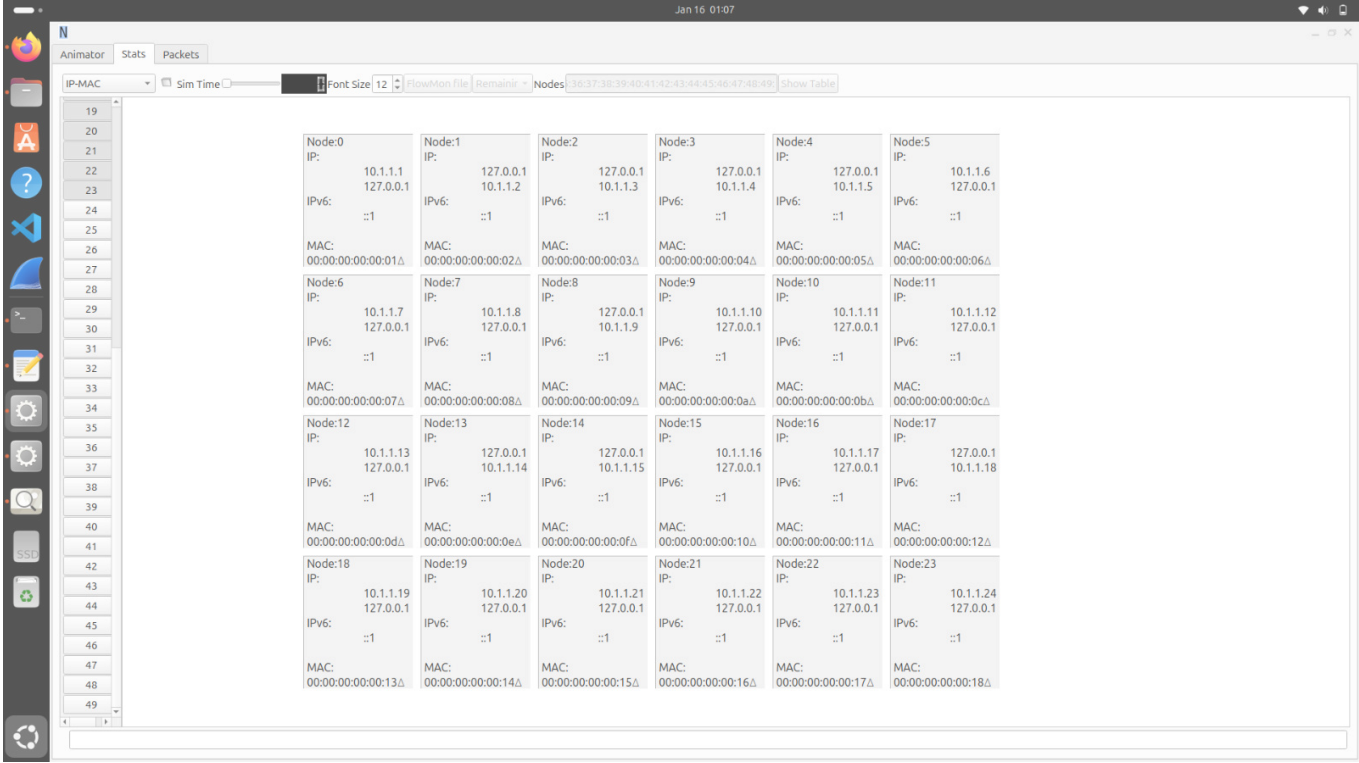


Fig. 10 Flow-monitor statistics from a network simulation

8.3 NetAnim Visualization of Attack and Mitigation

NetAnim is a visualization tool for an NS-3 network simulator that graphically represents simulation nodes, their traffic, and properties (such as IP and MAC addresses). It is used to analyze network activity and performance. The **figure 11** displays node IDs, IPv4/IPv6, and MAC address mappings from a network simulation, aiding device identification and traffic analysis.



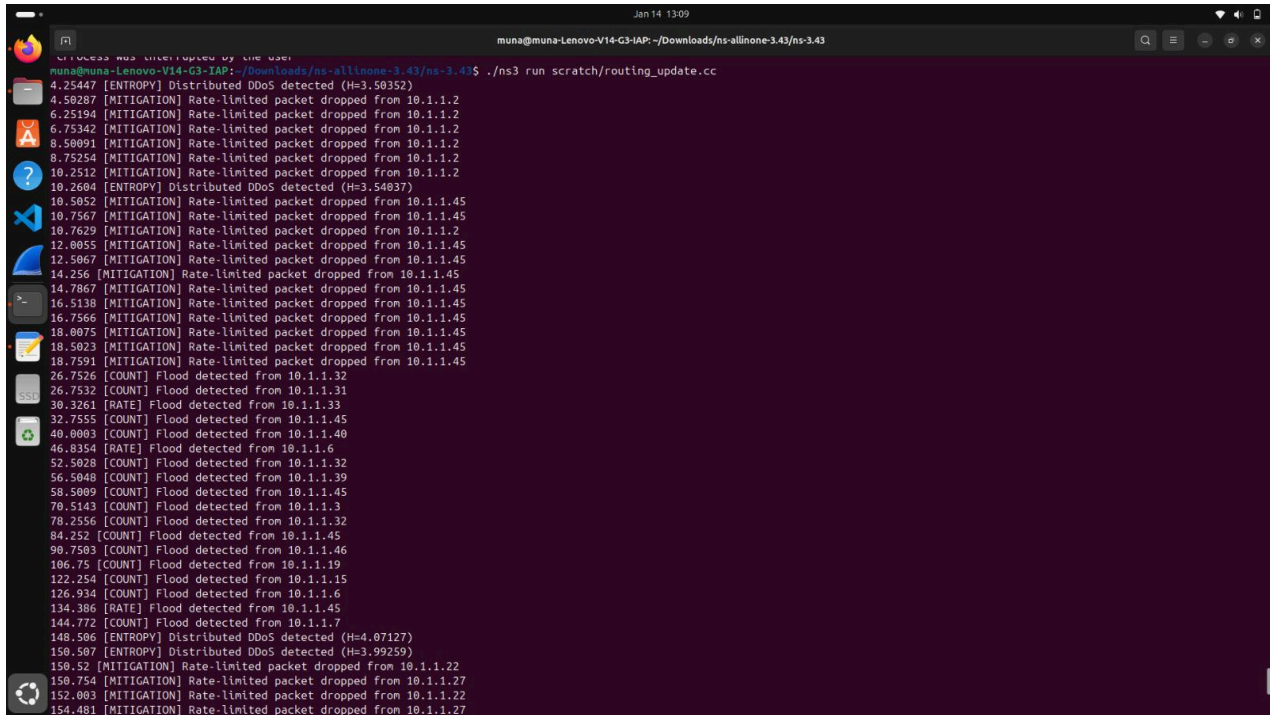
The screenshot shows the NetAnim application window with a table of node information. The table has columns for Node ID, IP, IPv6, and MAC. The nodes are listed in a grid format, with each node's details displayed in a separate row. The table is titled 'IP-MAC' and includes a 'Show Table' button. The nodes are numbered from 0 to 23, and their IP, IPv6, and MAC addresses are listed below each node ID.

Node	IP	IPv6	MAC
Node:0	10.1.1.1	::1	00:00:00:00:00:01
Node:1	127.0.0.1	::1	00:00:00:00:00:02
Node:2	127.0.0.1	::1	00:00:00:00:00:03
Node:3	127.0.0.1	::1	00:00:00:00:00:04
Node:4	127.0.0.1	::1	00:00:00:00:00:05
Node:5	10.1.1.6	::1	00:00:00:00:00:06
Node:6	10.1.1.7	::1	00:00:00:00:00:07
Node:7	10.1.1.8	::1	00:00:00:00:00:08
Node:8	127.0.0.1	::1	00:00:00:00:00:09
Node:9	10.1.1.10	::1	00:00:00:00:00:0a
Node:10	10.1.1.11	::1	00:00:00:00:00:0b
Node:11	10.1.1.12	::1	00:00:00:00:00:0c
Node:12	10.1.1.13	::1	00:00:00:00:00:0d
Node:13	127.0.0.1	::1	00:00:00:00:00:0e
Node:14	127.0.0.1	::1	00:00:00:00:00:0f
Node:15	10.1.1.16	::1	00:00:00:00:00:10
Node:16	10.1.1.17	::1	00:00:00:00:00:11
Node:17	127.0.0.1	::1	00:00:00:00:00:12
Node:18	10.1.1.19	::1	00:00:00:00:00:13
Node:19	10.1.1.20	::1	00:00:00:00:00:14
Node:20	10.1.1.21	::1	00:00:00:00:00:15
Node:21	10.1.1.22	::1	00:00:00:00:00:16
Node:22	10.1.1.23	::1	00:00:00:00:00:17
Node:23	10.1.1.24	::1	00:00:00:00:00:18

Fig. 11 IDs, IPv4/IPv6, and MAC address mappings from a network simulation

8.4 Successful Mitigation Scenario Demonstration

The successful operation of the mitigation mechanism is demonstrated through visual outputs and logged events. Detection alerts, trust value reductions, packet drops, and blacklist enforcement messages are observed in the simulation logs. the result show in the **fig 12**:



```
muna@muna-Lenovo-V14-G3-IAP: ~/Downloads/ns-allinone-3.43/ns-3.43$ ./ns3 run scratch/routing_update.cc
4.25447 [ENTROPY] Distributed DDoS detected (H=3.50352)
4.50287 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
6.25194 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
6.75342 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
8.50091 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
8.75254 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
10.2512 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
10.2604 [ENTROPY] Distributed DDoS detected (H=3.54037)
10.5052 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
10.7567 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
10.7629 [MITIGATION] Rate-limited packet dropped from 10.1.1.2
12.0055 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
12.5067 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
14.256 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
14.7867 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
16.5138 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
16.7566 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
18.0075 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
18.5023 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
18.7591 [MITIGATION] Rate-limited packet dropped from 10.1.1.45
26.7526 [COUNT] Flood detected from 10.1.1.32
26.7532 [COUNT] Flood detected from 10.1.1.31
30.3261 [RATE] Flood detected from 10.1.1.33
32.7555 [COUNT] Flood detected from 10.1.1.45
40.0003 [COUNT] Flood detected from 10.1.1.40
46.8354 [RATE] Flood detected from 10.1.1.6
52.5028 [COUNT] Flood detected from 10.1.1.32
56.5048 [COUNT] Flood detected from 10.1.1.39
58.5009 [COUNT] Flood detected from 10.1.1.45
70.5143 [COUNT] Flood detected from 10.1.1.3
78.2556 [COUNT] Flood detected from 10.1.1.32
84.252 [COUNT] Flood detected from 10.1.1.45
90.7503 [COUNT] Flood detected from 10.1.1.46
106.75 [COUNT] Flood detected from 10.1.1.19
122.254 [COUNT] Flood detected from 10.1.1.15
126.934 [COUNT] Flood detected from 10.1.1.6
134.386 [RATE] Flood detected from 10.1.1.45
144.772 [COUNT] Flood detected from 10.1.1.7
148.506 [ENTROPY] Distributed DDoS detected (H=4.07127)
150.507 [ENTROPY] Distributed DDoS detected (H=3.99259)
150.52 [MITIGATION] Rate-limited packet dropped from 10.1.1.22
150.754 [MITIGATION] Rate-limited packet dropped from 10.1.1.27
152.003 [MITIGATION] Rate-limited packet dropped from 10.1.1.22
154.481 [MITIGATION] Rate-limited packet dropped from 10.1.1.27
```

Fig. 12 Console output demonstrating successful DDoS mitigation

Chapter 9

Project Planning

9.1 Individual Tasks

Task	Description	Responsible Member(s)
Research and Literature Review	Study of MANETs, DDoS attacks, entropy-based detection techniques, trust models, and related work to identify research gaps and define project objectives.	Amna Zainab, Maryam Ilyas
Dataset Acquisition and Preprocessing	Analysis of network traffic generated through NS-3 simulations, including normal and attack traffic behavior for entropy and trust evaluation.	Amna Zainab
Model Development	Design and implementation of adaptive entropy-based DDoS detection and trust-based mitigation mechanism using NS-3.	Amna Zainab
Model Evaluation and Tuning	Performance evaluation using metrics such as throughput and packet delivery ratio; tuning of entropy thresholds and trust parameters.	Maryam Ilyas, Amna Zainab
System Integration and Testing	Integration of detection and mitigation modules with AODV routing protocol and testing under multiple simulation scenarios.	Amna Zainab
Documentation and Report Writing	Preparation of thesis chapters including implementation, results, visualization, and analysis.	Maryam Ilyas
Final Review and Presentation Preparation	Final proofreading, formatting, preparation of presentation slides, and viva preparation.	Maryam Ilyas

Table 2. Individual Task

9.2 Gantt Chart

Task	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
WP 1 – Proposal Submission										
WP 2 – Evaluation Phase										
WP 3 – Start implementation										
WP 4 – Project Write up										

Table 3. Gantt Chart

Chapter 10

Conclusion and Future Work

10.1 Conclusion

This project presented an **Adaptive Entropy Trust-Based DDoS Detection and Mitigation mechanism for Mobile Ad Hoc Networks (MANETs)**. The proposed system was designed and implemented using the NS-3 network simulator and integrated with the AODV routing protocol to address the growing threat of DDoS attacks in decentralized and dynamic MANET environments.

The detection framework combines count-based, rate-based, and entropy-based techniques to accurately identify both single-source and distributed DDoS attacks. To mitigate detected attacks, a trust-based approach was employed, where node behavior is continuously evaluated and mitigation actions such as probabilistic rate limiting and temporary blacklisting are applied adaptively.

Simulation results demonstrate that, without mitigation, DDoS attacks significantly degrade network performance in terms of throughput and packet delivery ratio. However, after applying the proposed mitigation mechanism, network performance improved considerably, confirming the effectiveness of adaptive trust-based control. The layered integration of detection and mitigation ensured reduced false positives while maintaining fairness for legitimate nodes.

Overall, the results validate that the proposed approach is **lightweight, adaptive, scalable, and effective** for protecting MANETs against DDoS attacks.

10.2 Future Work

Although the proposed system achieves effective DDoS detection and mitigation, several enhancements can be explored to further improve performance and adaptability.

AI-Based Adaptive Trust Learning

Future work can incorporate artificial intelligence techniques to dynamically learn trust patterns from network behavior. Machine learning models can analyze historical traffic data to predict malicious behavior more accurately and adjust trust values autonomously, reducing reliance on fixed thresholds.

Reinforcement Learning for Mitigation Decisions

Reinforcement Learning (RL) can be applied to optimize mitigation strategies by allowing the system to learn optimal actions through interaction with the network environment. By using reward-based learning, the mitigation mechanism can adaptively select the best response—such as rate limiting or blacklisting—based on network conditions and attack severity.

Real-Time Intrusion Prevention in MANETs

Future enhancements may focus on implementing the proposed mechanism in real-time MANET environments, such as vehicular ad hoc networks (VANETs) or military communication systems. Integration with real hardware and real-time monitoring tools would validate the practical applicability of the proposed approach beyond simulation.

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